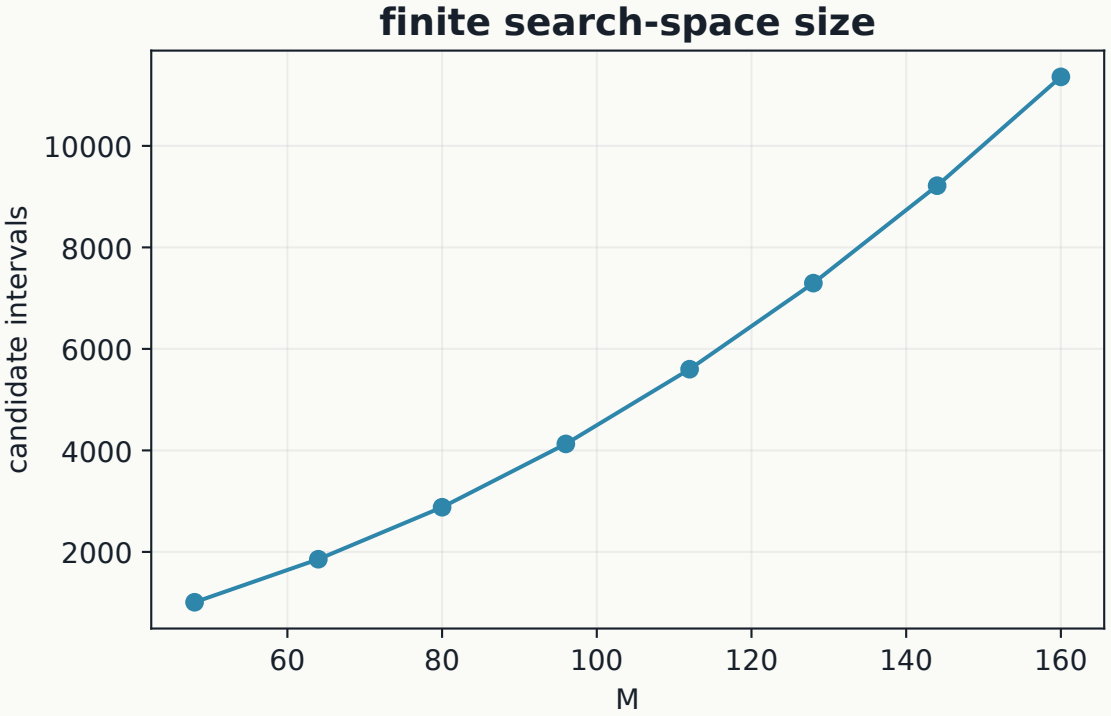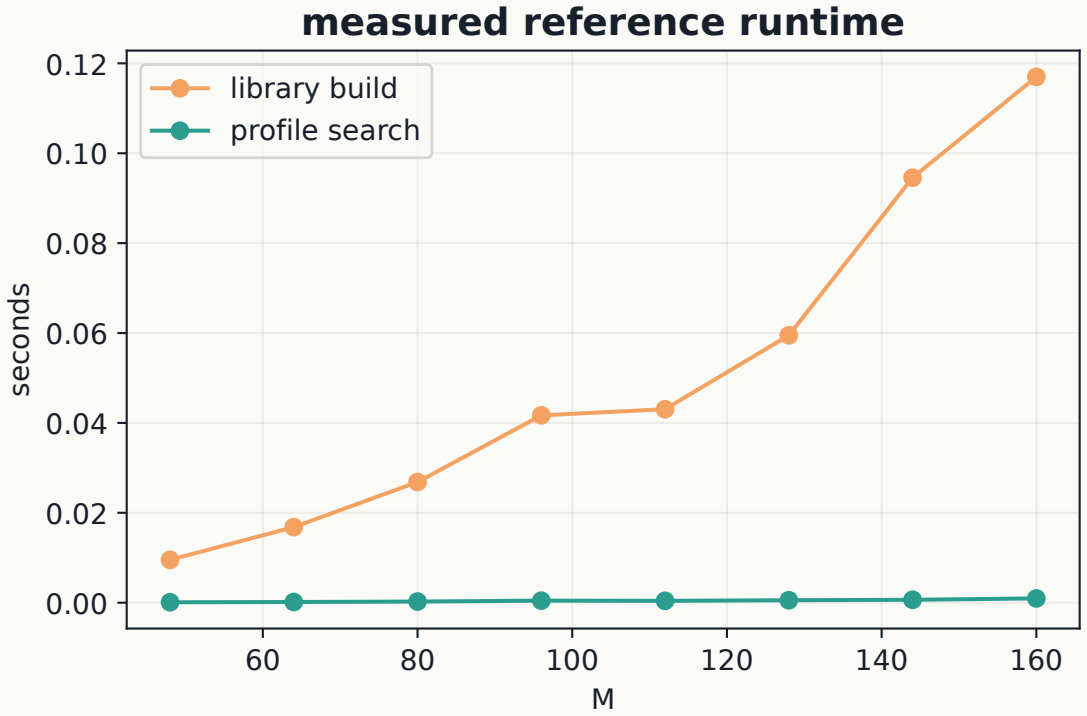The exhaustive solver checks every finite candidate. It is not claimed to be asymptotically optimal